

THE CLASS M GLOBAL TRIANGLE, WEIGHT-COMPLETED AND DS-REDUCED SCOPE

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ABSTRACT. The bulk–boundary–line triangle is proved for the standard chiral landscape via two compatible routes: (i) boundary–linear classes G, L, C hold at the chain level by direct bar–Hochschild identification; (ii) class M (Virasoro and principal $\mathcal{W}_N$) holds in the weight-completed category via the MC_5 weight-completed mechanism (Vol I ?? plus ??), and the chain-level bulk–boundary–line identification at class M is available in the principal and hook-type Drinfeld–Sokolov range through the DS–Hochschild compatibility theorem ??.

The direct-sum chain-level identification for class M at generic central charge is genuinely false ($S_4(\text{Vir}_c) = 10/[c(\zeta c + 22)] \neq 0$ forces bar cohomology in weight 4, obstructing direct-sum bar-cobar at chains); the correct theorem is in the weight-completed category. The surviving direct-sum curvature statement is a scalar central-channel statement in the completed curved bar model; it lives outside the weight-2 state space. The tempered-stratum theorem is restricted to non-logarithmic class M ; the logarithmic triplet $\mathcal{W}(p)$ remains open.

I. THE GLOBAL TRIANGLE

Theorem 1.1 (Global triangle for classes G, L, C). *For any chirally Koszul, boundary-linear logarithmic $SC^{\text{ch}, \text{top}}$ -algebra $\mathcal{A}$ of class G, L , or C in the standard landscape at non-critical level,*

$$\mathcal{A}_{\text{bulk}} \simeq Z_{\text{ch}}^{\text{der}}(\mathcal{B}_{\text{boundary}}) \simeq Z_{\text{ch}}^{\text{der}}(\mathcal{C}_{\text{line}}) \simeq \text{HH}_{\text{ch}}^*(\mathcal{A}_{\text{line}}^!)$$

holds at the chain level, together with the algebraic and geometric model equivalence for $\text{ChirHoch}^\bullet$ (geometric vs algebraic chiral-Hoch model compatibility).

Theorem 1.2 (Global triangle for class M , weight-completed). *For $\mathcal{A}$ in class M at generic central charge, equipped with the conformal-weight filtration $F^N \mathcal{A} = \bigoplus_{\Delta \geq N} \mathcal{A}_\Delta$, the global triangle holds in the weight-completed category: the Milnor- $\lim^1$ passage to $\widehat{\mathcal{A}} = \varprojlim_N \mathcal{A}/F^N \mathcal{A}$ assembles finite-weight stages, each of which satisfies the bar-cobar, HKR, and Hochschild identifications on its own. The chain-level identification at class M on the original complex requires the DS–Hochschild hypotheses stated below.*

2. WHY THE DIRECT-SUM CHAIN-LEVEL CLASS-M TRIANGLE IS FALSE

Proposition 2.1 (Direct-sum obstruction at weight 4). *Let $\mathcal{A} = \text{Vir}_c$ at generic c . The shadow coefficient*

$$S_4(\text{Vir}_c) = \frac{10}{c(\zeta c + 22)}$$

is nonzero for generic c ($S_2 = c/2$, $S_4 \neq 0$). By the Koszul-exponential universal-base theorem (Vol I ??), this nonvanishing shadow coefficient at weight 4 creates a bar cohomology class at total degree 2, weight 4, obstructing the direct-sum chain-level quasi-isomorphism between the bar-cobar composite and the identity. Hence the chain-level triangle at generic c fails in the direct-sum category. The correct scope is the weight-completed category.

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Proof. Explicit from the shadow recursion: at weight 4, the coefficient S_4 enters the bar-cobar obstruction class as the pairing with the $T \cdot T$ composite. Nonzero S_4 implies a nonzero class in $H^2(B^\Sigma(\text{Vir}_c))$ at weight 4. In the weight-completed category, the inverse system absorbs the obstruction through the completed finite-weight stages; at the direct-sum level, the class remains nonzero and obstructs quasi-isomorphism. $\square$

Remark 2.2 (Weight-completed scope). The obstruction belongs to the direct-sum ambient category. Completion replaces the isolated weight-4 obstruction by a compatible inverse system of finite-weight bar-cobar identifications. The direct-sum category has no such inverse-limit correction term.

3. THE SCALAR CURVATURE CHANNEL IN THE COMPLETED MODEL

Proposition 3.1 (Scalar central curvature channel). *Let $\mathcal{A}$ be a class- M chiral algebra on a smooth curve X with conformal vector T , at non-critical level, in the completed curved bar model. After projection to the scalar central channel, the genus- g curvature class is*

$$[d_{\text{fib}}^2]_{\text{scal}} = \kappa_{\text{ch}}(\mathcal{A}) \omega_g \otimes \mathbf{1}_{\mathcal{A}} \in H^2(\overline{\mathcal{M}}_{g,n}) \otimes Z(\mathcal{A})_0,$$

with ω_g the chosen Arakelov representative and $\kappa_{\text{ch}}(\mathcal{A})$ the chiral shadow coefficient of Vol I. The statement is a cohomological scalar-channel assertion; it does not identify a weight-2 state of $\mathcal{A}$ with the vacuum.

Proof. In the completed curved bar construction the genus- g square of the fibre differential is a degree-two class on the moduli factor with values in the central channel of the bar-Hochschild complex. Project to conformal weight zero. The centre in weight zero is $\mathbb{C}\mathbf{1}_{\mathcal{A}}$, so this projection is a scalar multiple of $\mathbf{1}_{\mathcal{A}}$. The Arakelov normalisation fixes the degree-two moduli representative as ω_g . Vol I identifies the scalar coefficient of this central channel with the chiral shadow coefficient $\kappa_{\text{ch}}(\mathcal{A})$. The argument uses the completed curved bar model and the scalar projection; it makes no claim about all higher-weight curvature components in the direct-sum complex. $\square$

Remark 3.2 (Direct-sum obstruction remains). The scalar-channel computation does not remove the weight-4 direct-sum obstruction of Proposition 2.1. It isolates the cohomological curvature coefficient that survives after completion. The full original-complex harmonic decomposition remains a separate chain-level problem.

Corollary 3.3 ($c_r = S_r$ at the level of shadow coefficients). *For $\mathcal{A}$ in class M , assume the harmonic factorisation hypothesis for the completed bar-Hochschild complex and the scalar-channel identification of Proposition 3.1 at weight r . Then the shadow coefficient $S_r(\mathcal{A})$ equals the corresponding harmonic coefficient $c_r(\mathcal{A})$:*

$$c_r(\mathcal{A}) = S_r(\mathcal{A}).$$

Proof. Under the harmonic factorisation hypothesis, the harmonic projector selects the scalar central channel at weight r . The shadow tower of Vol I computes the same scalar through the bar-intrinsic obstruction recursion. Hence the two coefficients agree. The harmonic factorisation input is the proof obligation; Hodge theory alone does not supply it. $\square$

4. THE COHOMOLOGICAL IDENTIFICATION (DS-HOCHSCHILD)

The cohomological reduction of Virasoro class M to class L is a relative DS comparison. The unreduced affine Hochschild complex has extra gauge directions.

Proposition 4.1 (DS-Hochschild relative comparison, cohomological). *Let $\mathcal{A}_L = V_k(\mathfrak{sl}_2)$ at non-critical level and $\mathcal{A}_M = \text{Vir}_c$ with central charge $c(k) = 1 - 6(k+1)^2/(k+2)$. The Drinfeld-Sokolov*

reduction $p: \mathcal{A}_L \rightarrow \mathcal{A}_M$ identifies the reduced DS-relative chiral Hochschild cohomology with the Virasoro chiral Hochschild cohomology. On unreduced cohomology the visible degrees are:

$$\begin{aligned} \text{ChirHoch}^0(\mathcal{A}_L) &= \mathbb{C} \xrightarrow{\sim} \mathbb{C} = \text{ChirHoch}^0(\mathcal{A}_M), \\ \text{ChirHoch}^1(\mathcal{A}_L) &= \mathfrak{sl}_2 \xrightarrow{\circ} \mathfrak{o} = \text{ChirHoch}^1(\mathcal{A}_M), \\ \text{ChirHoch}^2(\mathcal{A}_L) &= \mathbb{C} \cdot \Theta_k \xrightarrow{dc/dk} \mathbb{C} \cdot \Theta_c = \text{ChirHoch}^2(\mathcal{A}_M), \\ \text{ChirHoch}^{\geq 3}(\mathcal{A}_L) &= \mathfrak{o} \rightarrow \mathfrak{o} = \text{ChirHoch}^{\geq 3}(\mathcal{A}_M), \end{aligned}$$

with $dc/dk = -6(k+1)(k+3)/(k+2)^2 \neq 0$ at generic k .

Degree 1 is killed in the DS-relative complex by BRST exactness, inner absorption, and the screening identity (Vol I ??). Thus the arrow $\mathfrak{sl}_2 \rightarrow \mathfrak{o}$ is a quotient by gauge directions. Degree 2 is the chiral shadow depth class (Theorem H). Degrees ≥ 3 vanish by Theorem H concentration.

5. THE CHAIN-LEVEL IDENTIFICATION AT CLASS M

The chain-level lift of Proposition 4.1 is the DS–Hochschild comparison in Vol II ??. The statement is used only in its declared principal and hook-type range.

Theorem 5.1 (DS–Hochschild chain-level compatibility). *Let $\mathfrak{g}$ be a simple Lie algebra of classical type and $f \in \mathfrak{g}$ a principal or hook-type nilpotent. The DS strong deformation retract (p, i, h) induces a quasi-isomorphism of chiral Hochschild complexes*

$$(p \otimes p)^* : C_{\text{ch}}^\bullet(\mathcal{W}_k(\mathfrak{g}, f), \mathcal{W}_k(\mathfrak{g}, f)) \xrightarrow{\sim} C_{\text{ch}}^\bullet(V_k(\mathfrak{g}), V_k(\mathfrak{g}))^{Q_{\text{DS}}}$$

at the chain level, under the SDR and mixed-action hypotheses of Vol II ??, compatible with the E_3 -chiral action of Vol II ??. Combined with the class-L global triangle (Theorem 1.1), this gives the chain-level class-M global triangle for principal and hook-type nilpotents.

Corollary 5.2 (Class M chain-level bulk–boundary–line). *For $\mathcal{A}$ in class M with principal or hook-type Drinfeld–Sokolov presentation (Virasoro, principal $\mathcal{W}_N$, hook-type $\mathcal{W}_k(\mathfrak{g}, f)$) at generic central charge, and satisfying the DS–Hochschild hypotheses of Theorem 5.1, the full bulk–boundary–line global triangle of Theorem 1.1 holds at the chain level. For subregular and minimal nilpotents the theorem holds through the coset decomposition only when the same comparison hypotheses are verified for the coset factors; the weight-completed version of Theorem 1.2 holds in the completed ambient category.*

6. HARMONIC FACTORISATION

Conjecture 6.1 (Harmonic factorisation of bar Hochschild). *For $\mathcal{A}$ in the weight-completed class-M landscape at generic central charge, the harmonic projector on the bar-Hochschild complex $B^\Sigma(\mathcal{A})$ coincides at each weight with the symmetric-group averaging map $\text{av}: B^{\text{ord}}(\mathcal{A}) \rightarrow B^\Sigma(\mathcal{A})$ composed with the canonical inclusion:*

$$\mathcal{H} = \text{av} \circ \iota.$$

Under this hypothesis, the degree- r component of the harmonic projector computes the shadow coefficient $S_r(\mathcal{A})$, matching Corollary 3.3.

7. THE TEMPERED STRATUM, SCOPE-QUALIFIED

The non-logarithmic class-M tempered range is the declared generic or admissible exact-sector range below. Logarithmic triplet algebras remain outside the proved range: Gurarie 1993 and Flohr 1996 exhibit logarithmic amplitudes with unbounded Massey-product behaviour despite finite-dimensional Zhu algebras.

Theorem 7.1 (Tempered stratum, scope-qualified). *The tempered stratum of the class-M landscape contains:*

- (a) *Virasoro Vir_c at generic central charge $c \in \mathbb{C} \setminus \{\text{minimal series}\}$ in the completed exact-sector ambient (via $\rho_*(c) = c/6$);*
- (b) *Principal $\mathcal{W}_N$ for every $N \geq 2$ on the finite-envelope locus; the exact radius $\rho_*^{\mathcal{W}_N}(c) = |c|/[12(H_N - 1)]$ is available only under the harmonic-radius identification;*
- (c) *Irrational cosets $\text{Coset}(\mathfrak{g}, \mathfrak{h}, k)$ at generic level;*
- (d) *Non-logarithmic Schellekens VOAs in the standard central charge $c = 24$ catalogue.*

The logarithmic triplet algebras $\mathcal{W}(p)$, $p \geq 2$, are outside this theorem; their temperedness requires Adamović–Milas character-amplitude bounds.

Remark 7.2 (Non-tempered stratum on the non-logarithmic subset). The non-tempered stratum is empty on the non-logarithmic exact-sector standard landscape covered by Theorem 7.1. The C_2 -cofinite part is the admissible/minimal sublocus; generic Virasoro and generic cosets live in the completed generic ambient. The logarithmic $\mathcal{W}(p)$ subcase is open.

8. SUMMARY

- (1) **Classes G/L/C (chain level).** Direct bar-Hochschild identification; Theorem 1.1.
- (2) **Class M weight-completed.** MC_5 weight-completed mechanism via ?? + ??; Theorem 1.2. This is the weight-completed statement on the declared generic exact-sector class-M lanes; it is not a direct-sum chain-level theorem.
- (3) **Class M chain-level (principal / hook-type DS).** Vol II ?? bridges DS cohomology at chain level under its SDR and mixed-action hypotheses; Corollary 5.2.
- (4) **Scalar curvature channel.** The completed curved bar model has scalar central curvature $\kappa_{\text{ch}}(\mathcal{A})\omega_g \otimes \mathbf{1}_{\mathcal{A}}$ (Proposition 3.1); the harmonic projector equals averaging under the harmonic-factorisation hypothesis (Corollary 3.3).
- (5) **Direct-sum class-M genuinely false.** Nonzero $S_4(\text{Vir}_c)$ obstructs chain-level quasi-isomorphism at weight 4 in the direct-sum category (Proposition 2.1).
- (6) **Tempered stratum scope.** Virasoro, principal $\mathcal{W}_N$ finite-envelope loci, irrational cosets, non-logarithmic Schellekens; $\mathcal{W}(p)$ open (Theorem 7.1).

In the class-M sector, the bulk algebra of 3d quantum gravity is the chiral Hochschild cohomology of the boundary conformal algebra in the completed category. On the original complex this identification is the DS–Hochschild theorem under its stated hypotheses.

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